

Cells, Tissues & Body Systems

Your body has about 37 trillion cells. Each one is a tiny machine with a job. Cells build tissues, organs, and systems that keep you alive. This guided edition walks you through it step by step.

 Georgia Standard S7L2

WHAT YOU'LL LEARN

- | | |
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| 01 The Cell Is a System | 05 Levels of Organization |
| 02 Animal Cell Parts | 06 The Body's Systems |
| 03 Plant Cell Extras | 07 Systems Work Together |
| 04 Parts Work Together | 08 Review & Check |



CHAPTER WORD BANK

cell

organelle

nucleus

mitochondria

cell membrane

chloroplast

tissue

organ

organ system

organism

Look for the  boxes — they explain each word.

2.1 The Cell Is a System

A **cell** is the smallest living thing. A cell is not just a blob. It is a tiny **system** with many parts. Each part has a job.

cell = the smallest unit of life.

organelle = a tiny part inside a cell that has a job.

Think of a cell like a **factory**. A factory has an office, a power plant, and a cleanup crew. A cell has parts that do the same kinds of jobs. When all the parts work, the cell stays alive.

✓ Quick Check 2.1

1. What is the smallest unit of life?
2. What do we call a tiny part inside a cell that has a job?

2.2 Animal Cell Parts

Here are the main parts of an animal cell. Each one has a job.

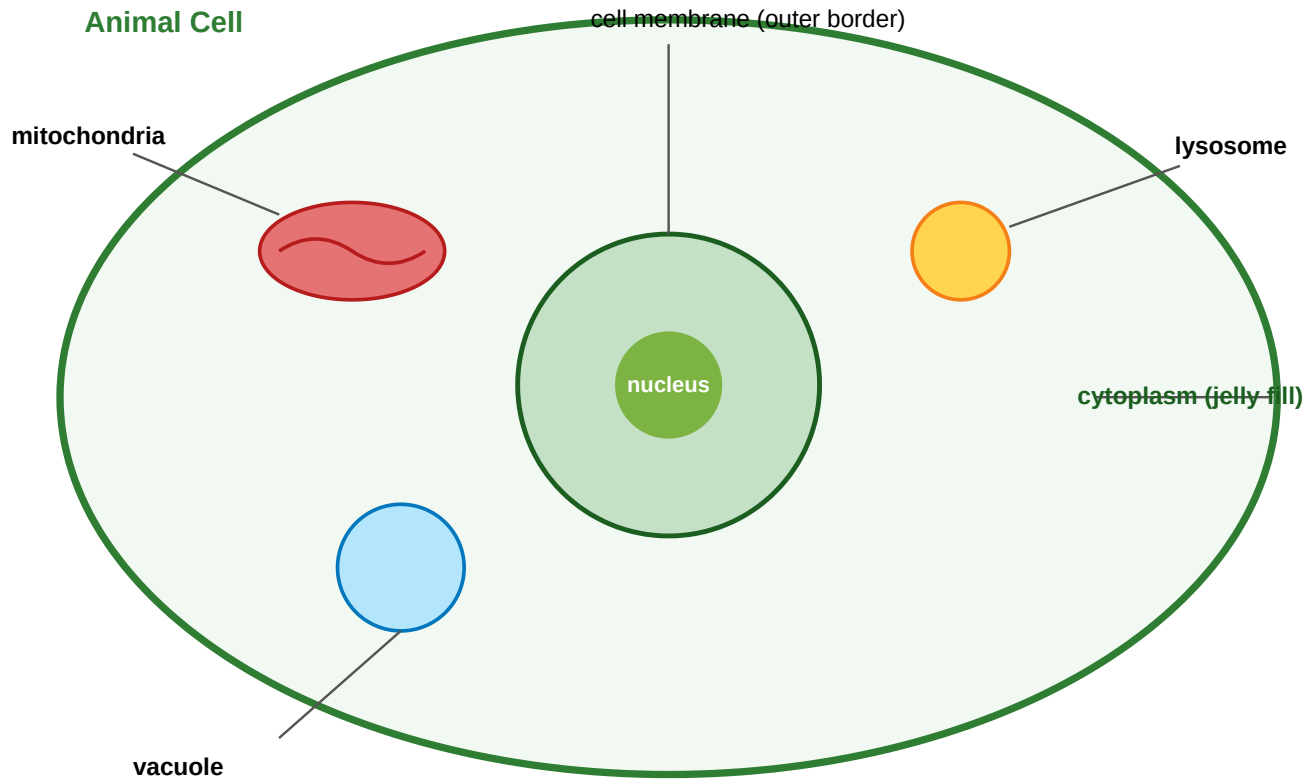


Figure 2.1 — An animal cell and its parts.

 Nucleus Boss; holds DNA	 Cytoplasm Jelly that fills the cell	 Cell Membrane Border; controls what goes in/out
 Mitochondria Makes energy from food	 Lysosome Cleans up waste	 Vacuole Storage

🔑 Match the Job

Boss = **nucleus**. Power plant = **mitochondria**. Door = **cell membrane**. Trash crew = **lysosome**. Storage closet = **vacuole**.

✓ Quick Check 2.2

1. Which part makes energy from food?
2. Which part is the "boss" and holds DNA?

2.3 Plant Cell Extras

Plant cells have all the animal cell parts — PLUS three extra parts. These extras help plants stand up and make food.

Plant Cell

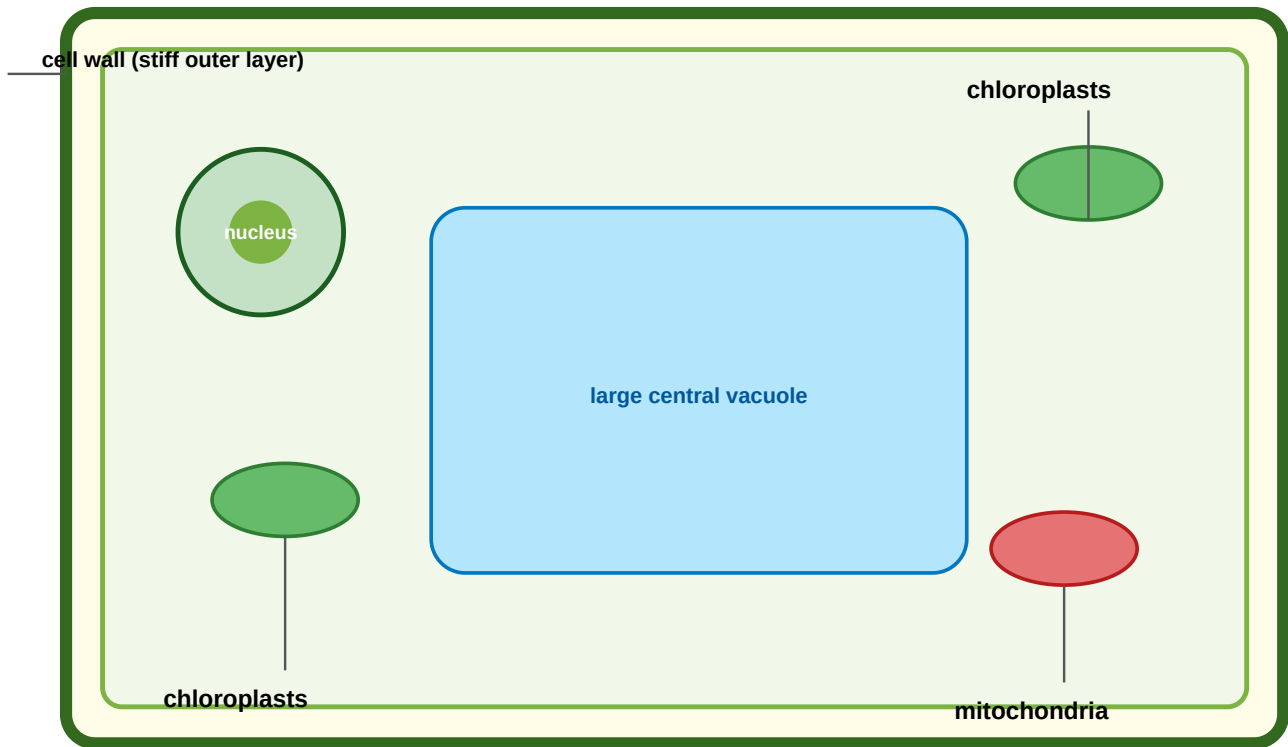


Figure 2.2 — A plant cell. See the cell wall, chloroplasts, and big vacuole.

Plant-Only Part	What It Does
Cell Wall	A stiff outer layer. Helps the plant stand up straight.
Chloroplasts	Green parts that catch sunlight and make food.
Big Vacuole	A large water storage sac. Keeps the plant firm.

★ Both vs. Plant-Only

Both cells have: nucleus, cytoplasm, membrane, mitochondria, lysosome.

Only plants have: cell wall, chloroplasts, big vacuole.

🔪 Sentence Starter

One part a plant cell has that an animal cell does NOT have is _____.

2.4 Parts Work Together

No part works alone. The parts work as a **team**. Here is how a cell gets energy:

★ Teamwork: Getting Energy (4 steps)

- **Step 1:** The **cell membrane** lets food in.
- **Step 2:** The **cytoplasm** carries the food.
- **Step 3:** The **mitochondria** make energy from the food.
- **Step 4:** The **lysosome** cleans up the waste.

✓ Quick Check 2.4

1. What part lets food into the cell first?
2. What part makes the energy?

2.5 Levels of Organization

One cell cannot run a whole body. Living things are built in **levels**. Small parts join to make bigger parts.

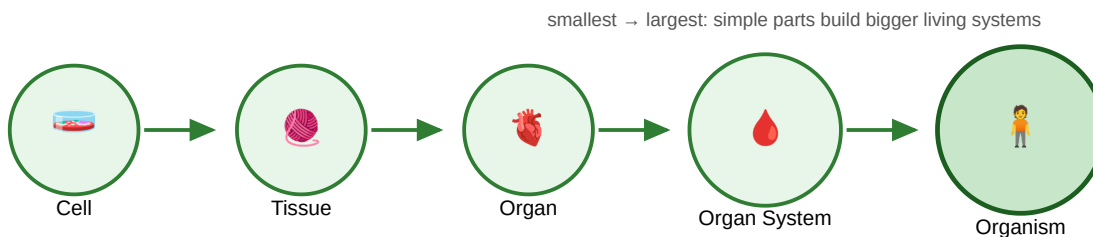


Figure 2.3 — From one cell up to a whole organism.

tissue = a group of similar cells.

organ = different tissues working together (like the heart).

organ system = a group of organs with one job.

organism = a whole living thing.

📖 FILL IN THE LEVELS

Put these in order from smallest to biggest:

C _____ → T _____ → O _____ → O _____ S _____ → O _____

2.6 The Body's Systems

Your body has several big systems. Each one has a job.

System	Its Job
Cardiovascular	Pumps blood (heart)
Respiratory	Breathing — takes in oxygen (lungs)
Digestive	Breaks down food (stomach)
Excretory	Removes liquid waste (kidneys)
Muscular	Moves the body (muscles)
Nervous	Sends signals (brain)
Immune	Fights germs

✓ Quick Check 2.6

1. Which system breathes and takes in oxygen?
2. Which system fights germs?

2.7 Systems Work Together

Systems do not work alone. To do anything — like running — many systems work together at the same time.

★ When You Run

- **Respiratory** system takes in air.
- **Cardiovascular** system pumps blood to your muscles.
- **Muscular** system moves your legs.
- **Nervous** system controls it all.

✍ Sentence Starter

When you run, the respiratory and cardiovascular systems work together to _____.

2.8 Review & Check

Great job! Here is everything in one place.

Big Idea	Remember This
Cell = system	Parts work together so the cell can live.
Animal parts	Nucleus, cytoplasm, membrane, mitochondria, lysosome, vacuole.
Plant extras	Cell wall, chloroplasts, big vacuole.
Levels	Cell → Tissue → Organ → Organ System → Organism.
Body systems	Cardiovascular, Respiratory, Digestive, Excretory, Muscular, Nervous, Immune.
Teamwork	Systems work together to keep you alive.

★ Standard Check — S7L2

You can now **(a)** show how cell parts work as a system, **(b)** put cells, tissues, organs, systems, and organisms in order, and **(c)** explain how body systems work together. 🎉